



HOW TO FIND THE RIGHT CAREER PATH

A PANEL OF REU ALUMNI

JUNE 25TH, 2025

MEET THE REU ALUMNI PANEL

Melissa Boswell, PhD

Zach Hostetler, PhD

Grace Krell, BS, MBA

Hannah Weinbaum, BS

Elsa King, BS

Molly Ehrig, BM

Samantha Schnell, BS

Lauren Washco, BS

MELISSA BOSWELL, PHD

- **Education & Employment**

- 2015 – B.S. Biomedical Engineering, University of Akron
- 2022 – MS & PhD, Bioengineering, Stanford University
- 2023 – Present: VP of Science and Research, Joe Gibbs Human Performance Institute

- **REU Class of 2015 - Advisor: Dr. Scott Gayzik**

- **REU Project:** Enhancing The Computational Human Body Models by Implementing Improved Neck Musculature and Head Kinematics and Conducting Full-body Simulations to Analyze Injury Biomechanics Across Various Crash Test Conditions

- **How did the REU help you find your career path?**

The REU program provided my first exposure to an academic research environment, allowing me to work alongside graduate students and gain valuable insight into their daily work and career trajectories. This experience deepened my passion for biomechanics and, while I didn't continue specifically with simulations, the knowledge gained became foundational to my work, including in my company's current work that combines simulation with experimental and applied approaches. Additionally, my collaboration with Dr. Gayzik on a successful NSF Graduate Research Fellowship Program application proved pivotal in securing admission to my preferred graduate program, effectively shaping my academic and professional path forward.



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ZACH HOSTETLER, PhD



■ Education & Employment

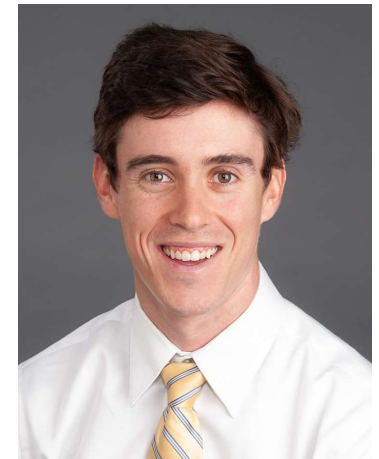
- 2017 - B.S. Mechanical Engineering, Baylor University
- 2019 – M.S. Biomedical Engineering, Virginia Tech - Wake Forest University SBES
- 2022 – Ph.D. Biomedical Engineering, Virginia Tech - Wake Forest University SBES
- Senior Engineer, Elemance (start up company) – eREU KEEN Program Mentor

- **Bio:** Zach Hostetler is a Senior Engineer at Elemance, a startup company based on his graduate research lab. He began his career as a 2017 summer intern at the Wake Forest Center for Injury Biomechanics, where he later earned his MS and PhD in computational injury biomechanics. His research focused on CT image analysis, human body modeling, and injury risk curve development for military and automotive safety. At Elemance, Zach combines software development with consulting for industry clients.

- **REU Class: 2017, Advisor(s): Dr. Ashley Weaver & Dr. Scott Gayzik**

- **How did the REU help you find your career path?**

While my summer internship affirmed my desire to study injury biomechanics in graduate school, it was also instrumental in guiding me through the application process. Not only did I have my summer mentor write a recommendation letter, but the graduate students I worked with also had great insight into programs I would be interested in and what to look for in a graduate school/advisor. Additionally, I had excellent summer research mentors that motivated me to be a strong mentor for undergraduate and younger graduate students during my time in graduate school.



SCHOOL OF BIOMEDICAL ENGINEERING AND SCIENCES

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GRACE KRELL, BS, MBA

■ Education & Employment

- 2018 – Founder & CEO of SweetSip
- 2022 – B.S. General Engineering with Double Concentrations in Biomedical and Industrial & Systems, East Carolina University
- 2024 – Master of Business Administration, Wake Forest University
- 2022 – Present: Senior Certification Engineer , Collins Aerospace

■ REU Class of 2021

■ Advisor: Dr. Ashley Weaver

■ REU Project: Sex Differences in Thoracic Injuries for Frontal Impact Motor Vehicle Occupants

■ How did the REU help you find your career path?:

Participating in the REU was a pivotal experience that helped me clarify my career path. At the time, I was uncertain whether biomedical engineering was the right concentration for me. However, once I began working on the CIREN project, I discovered a strong interest in vehicle crash safety and injury biomechanics. This hands-on experience ultimately led me to a position at Collins Aerospace, where I now certify aircraft seating systems. My work involves developing dynamic test plans, conducting crash tests using ATDs (Anthropomorphic Test Devices), and analyzing injury outcomes. Based on these assessments, we determine whether structural modifications are necessary to ensure the safety of all passengers.



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Collins Aerospace

A United Technologies Company



WAKE FOREST
UNIVERSITY



HANNAH WEINBAUM, BS

- **Education & Employment**

- 2024 – B.S. Mechanical Engineering, George Mason University
- 2024 – Present: Computational Analyst at Corvid Technologies, Missile Defense Systems (MDS)

- **REU Class of 2022 - Advisors: Dr. Joel Stitzel and Dr. Ashley Weaver**

- **REU Project:** Sex differences in thoracic injury causations among first row, frontal motor vehicle collision occupants

- **How did the REU help you find your career path?**

The REU program helped me decide my path for what I wanted to do after I finished undergrad. I went into the REU lost in my career path, not sure if I wanted to pursue a masters, PhD, or go straight into industry. This internship exposed me to both academia and industry, allowing me to see all possible paths before I graduated. Working with the Crash Injury Research and Engineering Network (CIREN) for my project, I learned a lot about vehicle safety and injury prevention and later found out this is a great passion of mine. After completing the REU with greater interest in vehicle safety, I was able to work for the Center for Collision Safety and Analysis at George Mason to develop a finite element model of a new vehicle that would be used in crash testing, and later work with Elemance to do quasi-static testing of rear-impact truck guards. I was able to continue developing my skills in finite element modeling, leading me to my current career where I do finite element modeling and simulations with an in-house high-fidelity computational physics solver for the missile defense system team here at Corvid.



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ELSA KING, BS

- **Education & Employment**

- 2024 – B.S. Biomedical Engineering, Rose-Hulman Institute of Technology
- 2024 – Present: PhD Student, Virginia Tech - Wake Forest University SBES

- **REU Class of 2022 - Advisors: Dr Weaver and Dr. Gayzik**

- **REU Project:** Finite Element Modeling of the Proximal Femur

- **How did the REU help you find your career path?**

The REU program confirmed my interest in research and introduced me to Wake Forest Institute for Regenerative Medicine! This allowed me to become a WFIRM summer scholar the following year and pursue my interest in Regenerative Medicine. I am currently a first-year PhD student under Dr. Anthony Atala working on renal bioprinting. The connections I made during my REU have helped me begin my journey into becoming a better researcher!



ROSE-HULMAN
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SCHOOL OF BIOMEDICAL ENGINEERING AND SCIENCES

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MOLLY EHRIG, MB

- **Education & Employment**

- 2022 – B.S. in Biology, minor in Statistics from Penn State University
- 2024 – Master of Biostatistics (MB) from Duke University
- 2024 – Present: Biostatistician II at Wake Forest University School of Medicine

- **REU Class of 2023 - Advisors: Drs. Jaime Speiser and Garrett Bullock**

- **REU Project:** - Imputation and missing indicators for handling missing longitudinal data: A simulation study based on electronic health record data

- **How did the REU help you find your career path?**

During the REU, I had the opportunity to learn about the topic of missing data while developing crucial coding, writing, and communication skills. My advisors introduced me to multiple different people in the Department of Biostatistics and Data Science, allowing me to learn about the department and the different types of work people in the department are involved in. My advisor sent me a job posting for a position here the fall after the REU, and in addition to reviewing my resume and cover letter she was one of my references. Ultimately, the REU set me on my career path at Wake!



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SAMANTHA SCHNELL, BS

- **Education & Employment**

- 2024 – B.S. Computer Science, University of North Carolina at Greensboro (UNCG)
- 2024 – Present: Masters in Analytics and Informatics, University of North Carolina at Greensboro (UNCG)
- Computer Science Research Assistant, University of North Carolina at Greensboro (UNCG)
- Data Science and Biostatistics Intern, Wake Forest School of Medicine

- **REU Class of 2023 - Advisor: Dr. Da Ma**

- **REU Project:** Alzheimer's Disease (AD) research involving genetics and extended existing research to create prediction model programs to predict the diagnosis of AD in patients based on their own genetic sequence using neural network/deep learning models and analysis

- **How did the REU help you find your career path?**

This internship allowed me to explore how computer science intersects with medicine and health research. With a background in biotechnology and a strong interest in genetics and health data, I was excited to apply my education in a new way. At the time, I was unsure of my next step—until I received an email about the REU Biomedical Informatics/CALIBIR internship. Through this experience, I delved into Alzheimer's research, genetic analysis, and discovered the power of machine learning, deep learning, and bioinformatics in medical research. It inspired me to pursue AI, especially NLP and ML, leading to my current roles as a research assistant at UNCG and a Data Science/Biostatistics intern at Wake Forest. My mentor, Dr. Ma, played a pivotal role in guiding me and showing me the broader applications of computer science. His support, along with the REU experience, helped shape my path in data analytics and AI. You never know where exploration will lead—but it's always worth the journey!



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LAUREN WASHCO, BS

- **Education & Employment**

- 2024 – B.S. Biochemistry, Muhlenberg College
- 2024 – Present: PhD Molecular Medicine and Translational Science, Wake Forest University School of Medicine

- **REU Class of 2023 - Advisor: Dr. Adam Hall**

- **REU Project:** Application of Solid-State Nanopores Fabricated by Automated Controlled Breakdown to Hyaluronan Analysis

- **How did the REU help you find your career path?**

The REU program provided the research experience that solidified my desire to pursue a PhD. This experience also opened my eyes to the field of Biomedical Engineering and I was able to see its overlap with biochemistry. Participating in an REU program at a medical institution also provided me with exposure to research translatable to human health and disease. The experience I had with the Hall lab throughout my time here was incredible. This REU led to my decision to pursue my PhD at Wake Forest under the guidance of Dr. Hall using solid-state nanopores to look at DNA damage in response to radiation exposure. I still am in contact with many of the other interns from my cohort and am roommates with another intern who also came to Wake Forest!



Wake Forest University
School of Medicine



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QUESTIONS
FOR OUR
PANELISTS:



CAREER PATH DISCOVERY

1. How did you figure out what career path was right for you?
2. Were there any moments of doubt or redirection in your journey? How did you handle them?
3. What factors were most important to you when choosing your first job after college?
4. Did your career goals change after your internship experience? If so, how?
5. What advice would you give to someone who's still unsure about what they want to do after graduation?

IMPACT OF THE SUMMER INTERNSHIP PROGRAM

1. How did the summer internship program influence your career decisions?
2. What skills or experiences from the internship helped you most in your current role?
3. Were there any mentors or connections from the program that played a role in your career journey?
4. Did the internship help you discover what you didn't want to do?
5. Can you share a specific project or moment from your internship that shaped your professional path?

PRACTICAL ADVICE FOR INTERNS

1. What should current interns focus on during their time in the program to get the most out of it?
2. How can interns turn this experience into a stepping-stone for future opportunities?
3. What would you do differently if you could go back to your internship summer?
4. How did you stay connected with people you met during the internship?
5. What's one thing you wish someone had told you when you were in our shoes?